Bagging and Random Forests: Bootstrap, Averaging, Variance Floor, OOB Error

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia Sources: `Lectures/L12_student_trees_bagging_boosting_worked_deck.pdf` (Part 2, slides 22–41), `Lectures/L12c_student_bagging_variance_bias_notes.pdf` (16-page notes: bagging regression trees, bias–variance), `Lectures/L12d_student_bagging_worked_example_on_trees.pdf` (7-page worked example), `Lectures/L12g_student_practice_trees_bagging_forests.pdf` (Parts B–D) · Previous: Decision trees · Next: Gradient boosting

Lecture 12 files: L12a Trees → **L12b Bagging & random forests** → L12c Gradient boosting → L12d AdaBoost → L12e XGBoost & comparison

> **How to use this file.** L12a ended with a problem: one deep tree changes completely when the data change a little (**high variance**). Bagging = **b**ootstrap **agg**regat**ing**: (1) make many training sets from the one you have by resampling, (2) grow a full tree on each, (3) average them. This file explains *why* it works with one formula, $\mathrm{Var}(\bar{T}) = \rho\sigma^2 + (1-\rho)\sigma^2/B$, and its consequence: a **floor** that only a **random forest** can lower. Along the way: what variance and bias actually mean, the 63% / 37% rule of the bootstrap, soft vs hard voting, and the free validation set called **out-of-bag error**. Part A is the maths step by step; Part B walks slides 22–41; Part B′ covers the two sets of notes; Part C is practice.

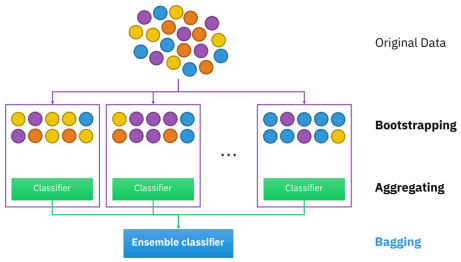
Two datasets are used:

- the **twelve runs** of L12a (classification; table in L12a), with new runs **A** (0.05, 32, no) and **B** (0.2, 64, yes);
- **eight points** for regression (notes L12c): x = model size (millions of parameters), y = minutes per epoch, made from a known truth $\mu(x) = x + 1$ plus noise:

i	1	2	3	4	5	6	7	8
x	1	2	3	4	5	6	7	8
y	2	3	3	5	6	8	8	9
$\mu(x)$	2	3	4	5	6	7	8	9
noise	0	0	−1	0	0	+1	0	0

What this file is really about:

$$\underbrace{T_b = \text{full tree on bootstrap sample } b}_{\text{low bias, high variance}} \longrightarrow \underbrace{\bar{T} = \tfrac{1}{B}\textstyle\sum_b T_b}_{\text{same bias}} \longrightarrow \underbrace{\operatorname{Var}(\bar{T}) = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{B}}_{\text{floor } \rho\sigma^2 \Rightarrow \text{ random forest lowers } \rho}$$



The whole idea in one picture: resample, train in parallel, combine. (Wikimedia Commons: Ensemble_Bagging.svg, Sirakorn, CC BY-SA 4.0)

Step-up path

You need. Variance, and how a single tree is built.

1. See that averaging independent predictions divides the variance by B.
2. Allow for correlation, which leaves a floor at $\rho\sigma^2$.
3. Make many training sets from one, with the bootstrap.
4. Lower ρ by hiding features at each split: a random forest.
5. Use the rows a tree never saw as a free validation set.

You can stop when you can explain what bagging fixes and what it cannot fix.

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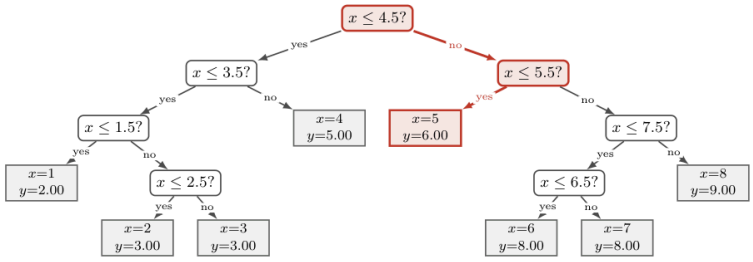
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Part A Maths From the Ground Up

A1 What "variance" and "bias" of a prediction mean

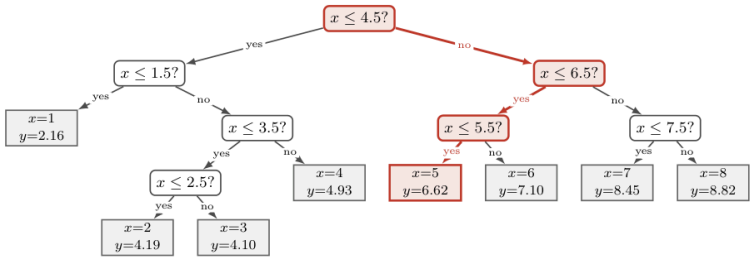
Step 1 — Fix one input x_0. A model's prediction there, $\hat{f}(x_0)$, is **one number**: follow the tree to a leaf and read it.

Step 2 — Imagine a different training set. Same x's, fresh noise. The tree would be different, x_0 would land in a different leaf, and the number would change.



Notes Fig. 1 — A fully grown regression tree on the eight points. In 1-D it predicts the y of the **nearest** training point (questions sit halfway between neighbours). $x_0 = 4.6 \to 6.00$ (truth 5.6). At $x_0 = 6.4$ it predicts 8 (truth 7.4): it copied the +1 noise of point 6.

Step 3 — Prediction variance = how much that one number moves **when only the training set changes**. It is *not* the scatter of y around the truth, and *not* how the prediction changes from one x to another.



Notes Fig. 2 — Same source, new noise ⇒ different questions, different leaf values ⇒ 5.45 vs 6.62 at the same x_0.

Step 4 — Measure it on six simulated training sets (prediction at $x_0 = 4.6$ = y at $x = 5$):

set	1	2	3	4	5	6	average
prediction	5.45	6.00	6.62	5.46	6.33	5.86	**5.953**
deviation	−0.50	+0.05	+0.67	−0.49	+0.38	−0.09	
squared	0.253	0.002	0.444	0.243	0.142	0.009	**0.182**

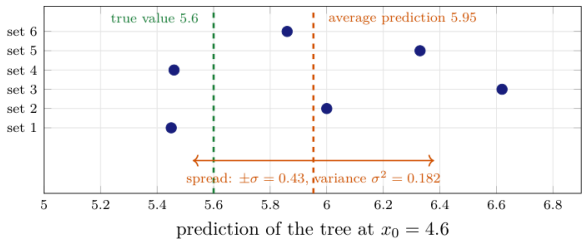
$$\mathrm{Var}(\hat{f}(x_0)) = E\big[(\hat{f}(x_0) - E\hat{f}(x_0))^2\big] \approx 0.182.$$

(Six sets are few; over many it settles near $\mathrm{Var}(\varepsilon) = 0.5^2 = 0.25$, because the tree simply copies y_5.)

Step 5 — Bias = how far the **average** prediction is from the truth:

$$\mathrm{Bias}(\hat{f}(x_0)) = E[\hat{f}(x_0)] - \mu(x_0) = E[y_5] - \mu(4.6) = 6 - 5.6 = 0.4.$$

A full tree has very low bias (it can follow any shape); the 0.4 is only because it must copy a neighbour.



Notes Fig. 3 — **Variance** = spread of the dots around their own average (orange). **Bias** = distance from that average to the truth (green).

A2 Bias² + variance + noise

Step 1 — Set-up. A fresh measurement $y_0 = \mu + \varepsilon_0$ with $E\varepsilon_0 = 0$, $\mathrm{Var}\,\varepsilon_0 = \sigma_\varepsilon^2$; a prediction $\hat{f}$ independent of ε_0, with mean $\bar{f} = E\hat{f}$.

Step 2 — Split the error into three pieces:

$$y_0 - \hat{f} = (\mu - \bar{f}) + (\bar{f} - \hat{f}) + \varepsilon_0.$$

Step 3 — Square and take expectations. The three squares give $(\mu - \bar{f})^2$, $E(\bar{f} - \hat{f})^2$, $E\varepsilon_0^2$. The three cross terms vanish: $E[\bar{f} - \hat{f}] = 0$, $E[\varepsilon_0] = 0$, and $\hat{f}$, ε_0 are independent with zero-mean factors. So

$$\boxed{E\left[(y_0 - \hat{f}(x_0))^2\right] = \text{Bias}^2 + \text{Var}(\hat{f}) + \sigma_\varepsilon^2}$$

Step 4 — Read it. Bias: model is systematically off. Variance: model moves with the sample. Noise: nobody can predict it. **We can only reduce the first two.** Bagging works on variance; boosting on bias (A12).

A3 Averaging independent predictions: σ²/B

Step 1 — B trees, each with variance σ^2, independent. Average $\bar{f} = \frac{1}{B}\sum_b \hat{f}_b$.

Step 2 — Two variance rules: $\text{Var}(cX) = c^2\text{Var}(X)$; for independent terms, $\text{Var}(\sum X_b) = \sum \text{Var}(X_b)$.

Step 3 — Combine:

$$\text{Var}(\bar{f}) = \frac{1}{B^2}\sum_b \text{Var}(\hat{f}_b) = \frac{B\sigma^2}{B^2} = \frac{\sigma^2}{B}.$$

Five independent trees with $\sigma^2 = 0.25$: $0.25/5 = 0.05$.

Step 4 — The bias does not move. $E[\bar{f}] = \frac{1}{B}\sum_b E[\hat{f}_b] = m(x)$, the same as one tree. **Averaging changes the variance and leaves the bias alone.**

A4 Correlated trees: the floor ρσ²

Trees grown on resamples of the **same** data look alike, so they are not independent.

Step 1 — Model it. Each tree has variance σ^2; every pair has correlation ρ, so $\text{Cov}(T_b, T_c) = \rho\sigma^2$ for $b \neq c$.

Step 2 — Pull out the constant: $\text{Var}(\bar{T}) = \frac{1}{B^2}\text{Var}\left(\sum_b T_b\right)$.

Step 3 — Expand the sum over all pairs: $\text{Var}\left(\sum_b T_b\right) = \sum_{b=1}^{B}\sum_{c=1}^{B}\text{Cov}(T_b, T_c)$.

Step 4 — Count the terms: B with $b = c$ (each σ^2) and $B(B-1)$ with $b \neq c$ (each $\rho\sigma^2$): total $B\sigma^2 + B(B-1)\rho\sigma^2$.

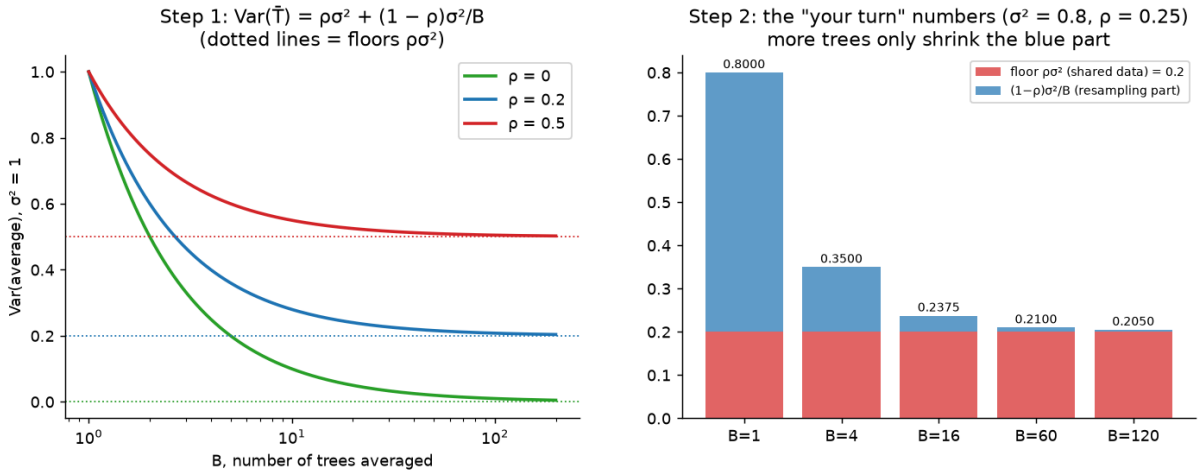
Step 5 — Divide by B^2:

$$\boxed{\mathrm{Var}(\bar{T}) = \frac{\sigma^2}{B} + \frac{B-1}{B}\rho\sigma^2 = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{B}}$$

Step 6 — Read it.

- $\rho = 0$: σ^2/B (independent trees).
- $\rho = 1$: σ^2 — averaging copies does nothing.
- $B \to \infty$: the **floor** $\rho\sigma^2$ remains, however many trees.

$\sigma^2 = 1$	ρ = 0	ρ = 0.3	ρ = 0.6	ρ = 1
B = 1	1.000	1.000	1.000	1.000
B = 4	0.250	0.475	0.700	1.000
B = 25	0.040	0.328	0.616	1.000
B = ∞	0.000	0.300	0.600	1.000



Step 1: curves flatten at their floor. Step 2: the "your turn" numbers ($\sigma^2 = 0.8$, $\rho = 0.25$): $B = 16$ gives $0.2 + 0.0375 = 0.2375$; to get within 0.01 of the floor 0.2 you need $0.6/B \le 0.01$, i.e. $B \ge 60$; doubling to 120 only goes 0.21 → 0.205.

What σ² and ρ mean at one input (deck slide 32). Fix run A; let $T_b = 1$ if tree b says D, 0 if C. If a share p of trees say D, $\sigma^2 = p(1-p)$:

share saying D, p	$\sigma^2 = p(1-p)$	meaning
1.00	0	every tree agrees: nothing to average away
0.872 (run A)	0.1115	most agree, a few differ
0.50	0.25	trees split evenly: one tree is no better than a guess

ρ = how much two trees' answers move together (0: one tells you nothing about the other; 1: always agree).

A5 The bootstrap: many training sets from one; the 63% rule

Step 1 — Draw n points with replacement from the n you have. A point may appear 0, 1, 2, 3… times. Same size, slightly different picture of the data.

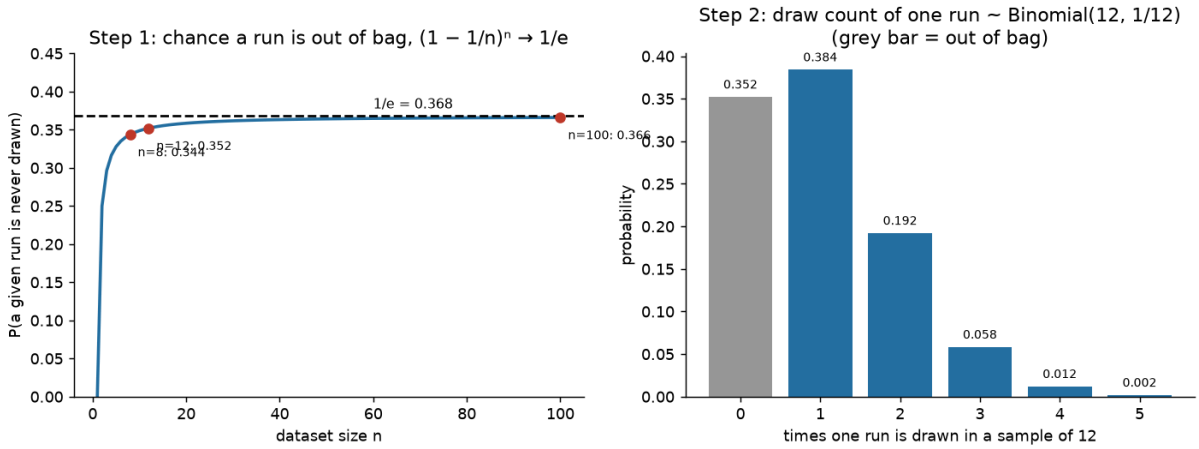
Step 2 — Chance a given point is never drawn. One draw misses it with probability $1 - \frac{1}{n}$; n independent draws all miss it with

$$\left(1 - \frac{1}{n}\right)^n \xrightarrow{n\to\infty} e^{-1} = 0.368.$$

Why the limit: $\ln(1-\frac{1}{n})^n = n\ln(1-\frac{1}{n}) \approx n(-\frac{1}{n}) = -1$.

n	8	12	20	100	1000	∞
P(left out)	0.344	0.352	0.358	0.366	0.368	0.368
share appearing	0.656	0.648	0.642	0.634	0.632	0.632

Step 3 — Rule of thumb: a bootstrap sample contains about **63%** of the distinct points; about **37%** are **out of bag**. That difference is what makes the trees differ.



Step 1: the out-of-bag chance converges to 1/e very fast. Step 2: one run's draw count in a sample of 12 — e.g. P(exactly twice) = $\binom{12}{2}(\frac{1}{12})^2(\frac{11}{12})^{10} = 0.192$.

A6 Bagging on the twelve runs, step by step

Step 1 — Draw three bootstrap samples (each cell: how many times the sample drew that run):

run	out	S1	S2	S3	run	out	S1	S2	S3
1	D	3	0	0	7	C	1	0	2
2	D	0	2	0	8	C	2	1	2
3	D	1	0	1	9	C	0	1	0
4	D	2	2	0	10	C	1	2	0
5	D	0	2	3	11	C	2	1	0
6	D	0	1	2	12	C	0	0	2

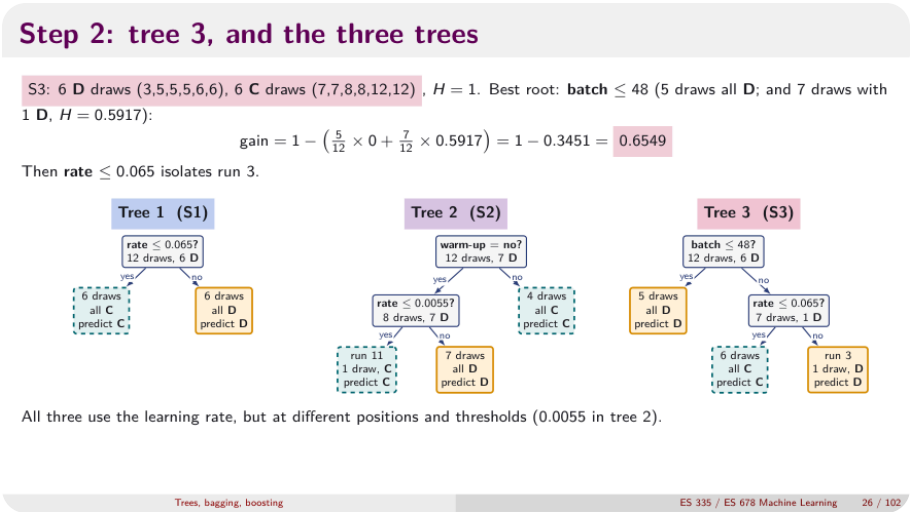
Column sums are 12. S1 missed 2, 5, 6, 9, 12; S2 missed 1, 3, 7, 12; S3 missed 1, 2, 4, 9, 10, 11.

Step 2 — Grow a full tree on each (same rule as L12a: largest entropy gain, stop when pure).

- **Tree 1 (S1):** 6 D draws (1,1,1,3,4,4), 6 C draws (7,8,8,10,11,11), $H = 1$. rate ≤ 0.065 splits them perfectly: gain $= 1 - 0 = 1.0000$ ⇒ stops after one question.
- **Tree 2 (S2):** 7 D (2,2,4,4,5,5,6), 5 C (8,9,10,10,11), $H(\frac{7}{12}) = 0.9799$. Best root **warm-up = no**: 8 draws with 7 D ($H(\frac{7}{8}) = 0.5436$) and 4 draws all C: $\text{gain} = 0.9799 - \frac{8}{12}(0.5436) = 0.9799 - 0.3624 = 0.6175$. Run 11 (C) is in the 8-draw group, so it asks rate ≤ 0.0055 to isolate it (gain 0.5436). *Why warm-up?*

S2 lacks runs 1 and 3 — the D runs **with** warm-up — so warm-up looks like the cause (rate ≤ 0.065 would only give 0.344).

- **Tree 3 (S3):** 6 D (3,5,5,5,6,6), 6 C (7,7,8,8,12,12), $H = 1$. Best root **batch ≤ 48**: 5 draws all D, 7 draws with 1 D ($H(\frac{1}{7}) = 0.5917$): $\text{gain} = 1 - \frac{7}{12}(0.5917) = 1 - 0.3451 = 0.6549$. Then rate ≤ 0.065 isolates run 3.



Slide 26 — Three samples of the same data ⇒ three different trees. All use the learning rate, but at different positions and thresholds.

Step 3 — Send a new run down every tree; read one number from each (T_b = fraction of D draws in the leaf reached):

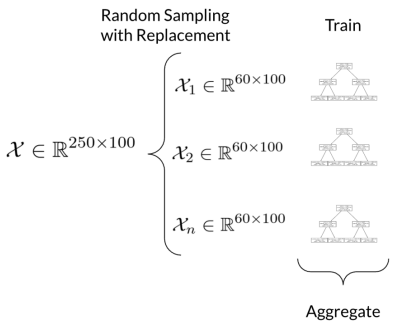
tree	run A (0.05, 32, no): leaf	$T_b(A)$	run B (0.2, 64, yes): leaf	$T_b(B)$
1	rate ≤ 0.065 yes → 0/6	**0**	rate ≤ 0.065 no → 6/6	**1**
2	warm-up no; rate > 0.0055 → 7/7	**1**	warm-up yes → 0/4	**0**
3	batch ≤ 48 yes → 5/5	**1**	batch > 48; rate > 0.065 → 1/1	**1**

Step 4 — Average, threshold at ½:

$$\bar{T}(A) = \tfrac{1}{3}(0+1+1) = \tfrac{2}{3} \Rightarrow \mathbf{D}, \qquad \bar{T}(B) = \tfrac{1}{3}(1+0+1) = \tfrac{2}{3} \Rightarrow \mathbf{D}.$$

Same answer as the single tree — but the single tree **showed no doubt**; the bag shows one tree in three disagrees. *Averaging turned a hidden disagreement into a number.*

Step 5 — Three trees give a rough estimate. With three trees $\bar{T} \in \{0, \frac{1}{3}, \frac{2}{3}, 1\}$. With 20,000 bootstrap trees (simulation) the share saying D settles at **0.872** (A) and **0.934** (B); with 200 trees, 0.845 and 0.965.



Resample → train → aggregate, in general form. (Wikimedia Commons: Random_Forest_Bagging_Illustration.png, Harry585, CC BY-SA 4.0)

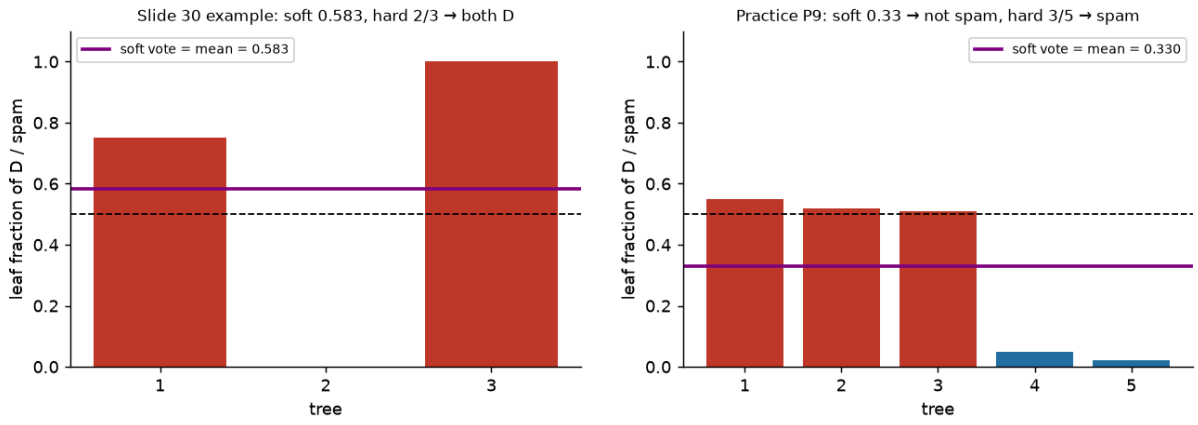
A7 Soft vote vs hard vote; why a vote helps

Step 1 — Two ways to combine.

- **Soft vote:** average the trees' leaf fractions (what A6 did).
- **Hard vote:** each tree casts D if its fraction > ½, else C; count.

Step 2 — Example (slides 30–31). Leaves contain (4 draws, 3 D), (5, 0 D), (2, 2 D): $T = 0.75, 0, 1$. Soft: $\frac{1}{3}(0.75 + 0 + 1) = 0.583 \Rightarrow$ D. Hard: D, C, D ⇒ $\frac{2}{3} = 0.667 \Rightarrow$ D. Same answer, different confidence.

Step 3 — They can disagree (practice P9): fractions 0.55, 0.52, 0.51, 0.05, 0.02. Soft: 0.33 ⇒ not spam. Hard: 3 of 5 ⇒ spam. Soft uses confidence (two very sure trees pull hard); hard counts every tree once. Soft is the usual default when trees give probabilities.

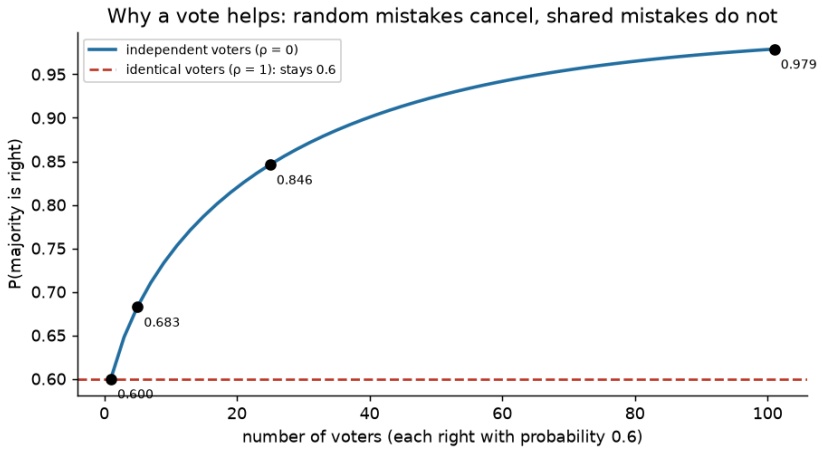


Left: the rules agree. Right: three borderline trees out-vote two confident ones under hard voting, but not under soft voting.

Step 4 — Why a vote helps. Each voter right with probability 0.6, independent:

voters	1	5	25	101
P(majority right)	0.600	0.683	0.846	0.979

(5 voters: $\sum_{k=3}^{5} \binom{5}{k} 0.6^k 0.4^{5-k} = 0.683$.) **Two warnings:** (1) if the voters make the same mistakes ($\rho = 1$) the majority is no better than one voter; (2) a vote cannot fix a **bias**: if every voter is wrong for the same systematic reason, so is the majority.



Random mistakes cancel in a large majority; shared mistakes do not.

A8 Bagging regression trees on the eight points

Step 1 — Five bootstrap samples (sorted):

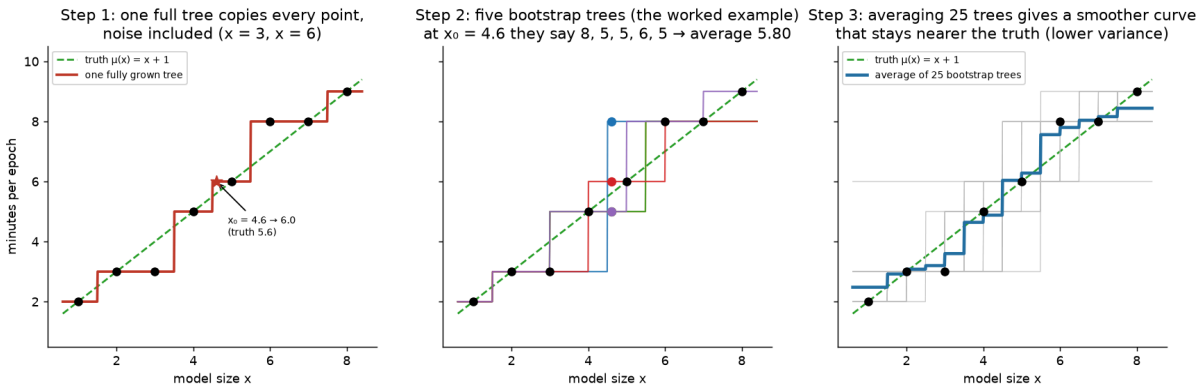
sample	points drawn	left out	distinct
1	1, 1, 2, 2, 3, 6, 6, 7	4, 5, 8	5
2	1, 2, 2, 2, 4, 4, 7, 7	3, 5, 6, 8	4
3	1, 1, 1, 2, 4, 4, 7, 7	3, 5, 6, 8	4
4	1, 2, 3, 3, 3, 5, 5, 7	4, 6, 8	5
5	1, 2, 2, 2, 4, 4, 6, 8	3, 5, 7	5

Step 2 — Grow a full tree on each. Each predicts the y of the nearest point **in its sample**. At $x_0 = 4.6$:

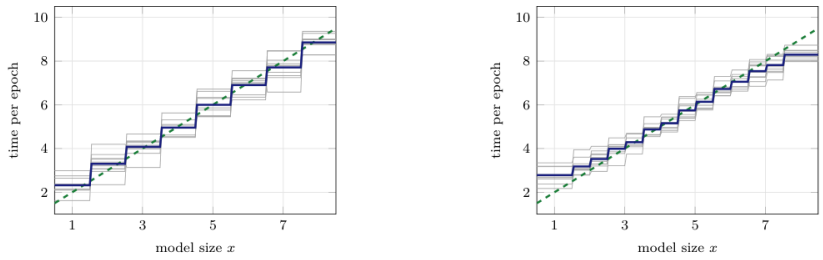
tree	distinct x in sample	nearest to 4.6	$\hat{f}_b$
1	1, 2, 3, 6, 7	6	8.0
2	1, 2, 4, 7	4	5.0
3	1, 2, 4, 7	4	5.0
4	1, 2, 3, 5, 7	5	6.0
5	1, 2, 4, 6, 8	4	5.0

Step 3 — Average: $\hat{f}_{\text{bag}}(4.6) = 29/5 = \mathbf{5.80}$ (single tree: 6.0; truth: 5.6).

Each bootstrap tree alone is *worse* than the tree on all eight points (it has seen fewer distinct points), but their errors partly cancel. **Why not train B trees on the same data?** They would be identical ($\rho = 1$).



Step 1 → Step 3: one tree, five trees, the average of 25.



Notes Fig. 4 — Low bias, high variance (left) vs bagged: **much lower variance** (right).

A9 Where ρ comes from; checking the formula

Step 1 — Check σ²/B by simulation (resample the twelve runs, grow a tree, record its answer at run A; $p = 0.872 \Rightarrow \sigma^2_{\text{boot}} = 0.1115$):

B	measured variance of the average	σ^2_{boot}/B
1	0.1118	0.1115
3	0.0362	0.0372
10	0.0111	0.0111
50	0.0021	0.0022

They agree — but **no floor**. Why? Here only the resampling varies; the dataset is fixed.

Step 2 — Split the randomness in two. Given the data $\mathcal{D}$, bootstrap trees are independent. Let $\mu(\mathcal{D}) = E[T \mid \mathcal{D}]$. By the laws of total covariance/variance, for $b \neq c$:

$$\text{Cov}(T_b, T_c) = \underbrace{E[\text{Cov}(T_b, T_c \mid \mathcal{D})]}_{=0} + \text{Var}(\mu(\mathcal{D})), \qquad \sigma^2 = \text{Var}(\mu(\mathcal{D})) + E[\text{Var}(T \mid \mathcal{D})].$$

Step 3 — Read off the two parts:

$$\rho\sigma^2 = \text{Var}(\mu(\mathcal{D})) \ \ (\text{due to the data}), \qquad (1-\rho)\sigma^2 = E[\text{Var}(T \mid \mathcal{D})] \ \ (\text{due to resampling}).$$

Averaging removes the resampling part, not the data part: that is the floor.

Step 4 — Numbers. On a simulated 200-run example: $\sigma^2 = 0.071$, $\rho = 0.26$, floor 0.019; a single full tree's validation error 0.144 fell to **0.115** for a 200-tree bag. On the eight-point regression (notes): one bootstrap tree has $\sigma^2 = 0.683$; solving the formula for ρ, $\rho = \dfrac{\text{Var}_B - \sigma^2/B}{\sigma^2 - \sigma^2/B} \approx 0.18$ at $B = 100$, and then the formula *predicts* the other rows:

B	1	5	10	25	100
simulated	0.683	0.237	0.181	0.146	0.131
$\rho\sigma^2 + (1-\rho)\sigma^2/B$	0.683	0.237	0.181	0.148	0.131

Step 5 — Measuring ρ directly needs many training sets (simulation): repeat K times "new training set → two bootstrap trees → their predictions u_k, v_k", then

$$r = \frac{\frac{1}{K}\sum(u_k - \bar{u})(v_k - \bar{v})}{\sqrt{\frac{1}{K}\sum(u_k - \bar{u})^2 \cdot \frac{1}{K}\sum(v_k - \bar{v})^2}}.$$

Six pairs at $x_0 = 4.6$ give $r = 0.062/\sqrt{0.299 \times 0.436} = 0.17$; 6000 pairs give 0.18. With real data there is one training set, so ρ is not measured directly; its effect shows as the OOB error **stopping improving** as B grows.

A10 Random forests: lower ρ by hiding features

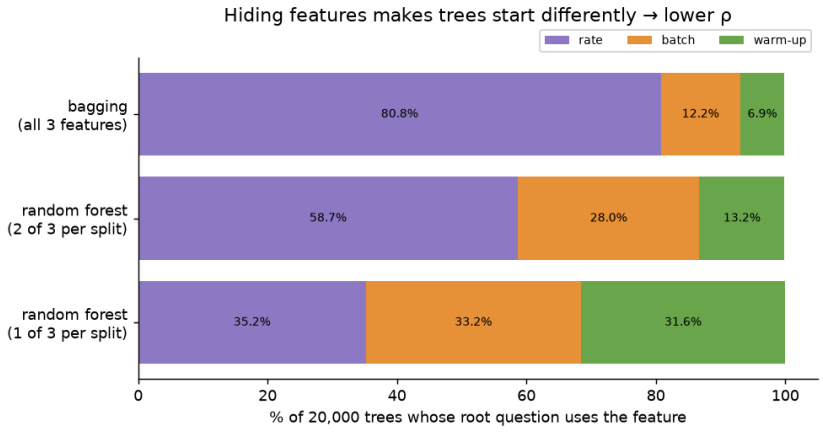
Step 1 — Are bagged trees really different? Not very. Root-question feature over 20,000 bootstrap trees:

	rate	batch	warm-up
bagging (all 3 features at every split)	**80.8%**	12.2%	6.9%
random forest, 2 random features per split	58.7%	28.0%	13.2%
random forest, 1 random feature per split	35.2%	33.2%	31.6%

Plain bagging starts with rate in four trees of five ⇒ trees alike ⇒ ρ high ⇒ floor high.

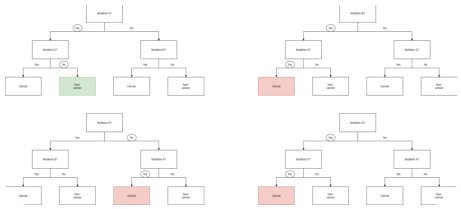
Step 2 — The random-forest trick. At **every split** offer only a random subset of m of the p features (often $m \approx \sqrt{p}$). Trees not offered the best feature must start elsewhere ⇒ they differ more ⇒ **ρ falls** ⇒ floor falls.

Step 3 — Where 58.7% comes from. With 2 of 3 offered, rate is offered at the root in $\frac{2}{3}$ of trees; when offered it wins 88% ⇒ $\frac{2}{3} \times 0.88 = 0.587$.



The fewer features offered, the more evenly trees start ⇒ less alike.

Step 4 — The price. Each tree is a little worse (larger σ²). Simulated: σ² 0.071 → 0.082, ρ 0.26 → 0.20, floor **0.019 → 0.016**. Usually the drop in ρ wins — but not always (practice P18): if hiding features makes trees too weak, bagging wins. m is a setting to tune.



Many different trees, one vote.
(Wikimedia Commons: Random_Forest_Diagram_with_Four_Trees.png, CollaborativeGeneticist, CC BY-SA 4.0)

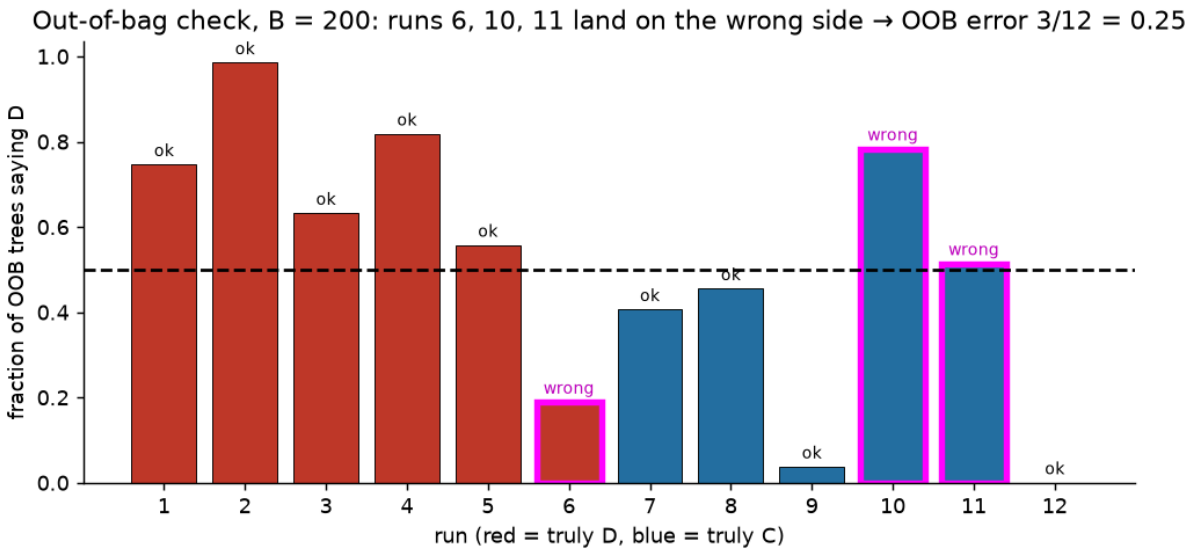
A11 Out-of-bag error: a free validation set

Step 1 — The idea. Each tree missed ≈37% of the runs. For one training run, let **only the trees that never saw it** vote, and compare with its true class.

Step 2 — The twelve runs, B = 200 (each run out of bag in ≈70 trees):

run	true	trees not drawing it	frac. D	OOB	right?
1	D	63	0.746	D	yes
2	D	65	0.985	D	yes
3	D	68	0.632	D	yes
4	D	77	0.818	D	yes
5	D	70	0.557	D	yes
6	D	74	0.189	C	**no**
7	C	69	0.406	C	yes
8	C	66	0.455	C	yes
9	C	80	0.037	C	yes
10	C	69	0.783	D	**no**
11	C	70	0.514	D	**no**
12	C	68	0.000	C	yes

Step 3 — OOB error = 3/12 = 0.25. The single tree scored on its own runs: 0 errors — an optimistic estimate. OOB is honest (no run is predicted by a tree that trained on it) and costs nothing: **no runs held out**.



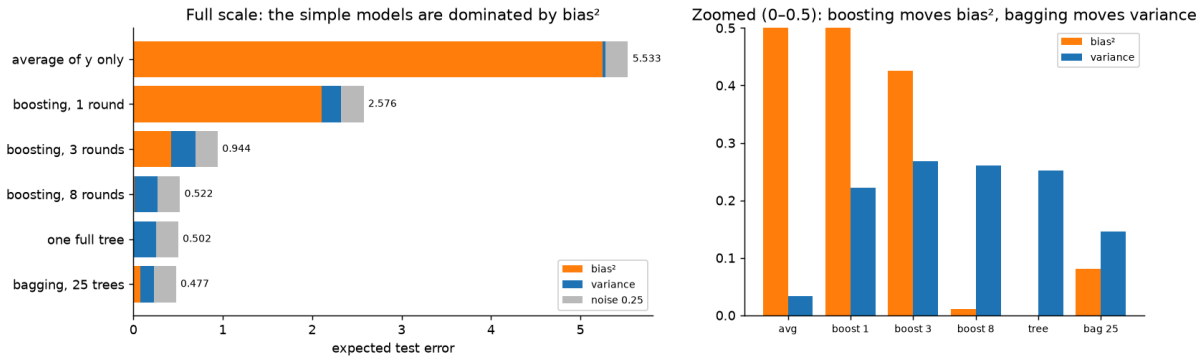
Runs 6, 10, 11 are the hard ones — exactly the low-rate/small-batch corner that needed the warm-up question.

Step 4 — Uses. (i) Estimate the error of a forest without a validation set. (ii) Choose B and m: pick the smallest OOB error. In scikit-learn: `RandomForestClassifier(oob_score=True)`.

A12 What bagging fixes and what boosting fixes

Step 1 — Measure all three error parts (4000 simulated training sets of eight points, truth $x + 1$, noise sd 0.5 $\Rightarrow \operatorname{Var} \varepsilon = 0.25$; averaged over the eight x):

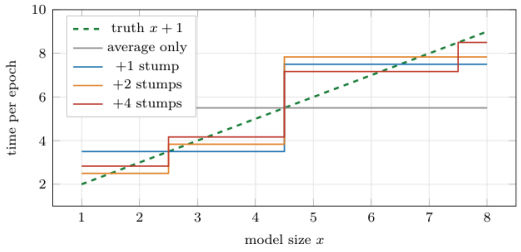
method	bias²	variance	noise	expected error
average of the y's only (no tree)	5.250	0.033	0.250	5.533
boosting, 1 round (ν = 0.5)	2.104	0.222	0.250	2.575
boosting, 3 rounds	0.426	0.268	0.250	0.944
boosting, 8 rounds	0.011	0.261	0.250	0.522
one fully grown tree	0.000	0.252	0.250	0.502
bagging, 25 fully grown trees	0.081	**0.146**	0.250	**0.477**



Read by columns: boosting drives **bias²** from 5.25 → 0.011; bagging cuts **variance** 0.252 → 0.146 (paying a little bias, 0.081, because bootstrap trees see fewer points).

Step 2 — How can a sum of weak stumps have low bias? Bias belongs to the **whole model**. Each stump is fitted to the **residual** (what is still wrong), and the model is the sum. One stump has 2 levels; a sum of t stumps can have up to $t + 1$ levels, so the family grows every round. Noise-free example (truth $x + 1$, full step):

round	stump added	F(1)	F(2)	F(3)	F(4)	F(5)	F(6)	F(7)	F(8)	bias²
0	start: average	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.50	5.25
1	x ≤ 4.5: −2.00, +2.00	3.50	3.50	3.50	3.50	7.50	7.50	7.50	7.50	1.25
2	x ≤ 2.5: −1.00, +0.33	2.50	2.50	3.83	3.83	7.83	7.83	7.83	7.83	0.92
3	x ≤ 7.5: −0.17, +1.17	2.33	2.33	3.67	3.67	7.67	7.67	7.67	9.00	0.72
4	x ≤ 4.5: +0.50, −0.50	2.83	2.83	4.17	4.17	7.17	7.17	7.17	8.50	0.47



Notes Fig. 6 — Weak pieces, strong sum. The full mechanism is L12c.

Step 3 — Side by side.

	bagging (random forest)	boosting
trees	deep: low bias, high variance	shallow: high bias, low variance
built	independently, on bootstrap samples (parallel)	one after another, on what is still wrong
combined by	average / vote	sum of small corrections
fixes	**variance**, down to the floor $\rho\sigma^2$	**bias**, step by step
more trees	never hurts; variance falls to a floor	too many rounds start to fit noise
main settings	number of trees; features per split	step size; rounds; depth

One sentence: *bagging averages away the variance of deep trees; boosting adds up small trees to remove the bias of shallow ones.*

Common confusions.

- *Why trees?* Deep trees are unstable — exactly what bagging fixes. Linear regression barely moves between samples, so bagging hardly helps it.
- *Does bagging overfit when B is large?* No: more trees lower variance toward the floor; the bias doesn't get worse.
- *Is the bootstrap cheating (points repeat)?* Repeats only change how much a point counts in one tree; the average over many resamples is used, and its error is checked out-of-bag.

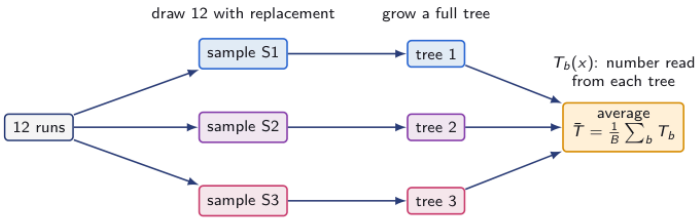
Part B Slide by Slide (slides 22–41)

Slide 22 Part 2: Bagging (divider)

Colours: tree 1 (S1), tree 2 (S2), tree 3 (S3).

Slide 23 Bagging: the procedure

Bagging: the procedure



Bagging mainly reduces **variance**: deep trees have low bias but change a lot with the data.

Slide 23 — Bagging mainly reduces **variance**: deep trees have low bias but change a lot with the data.

Slide 24 Step 1: draw three bootstrap samples

Step 1: draw three bootstrap samples

run	rate	batch	warm	out	S1	S2	S3
1	0.3	256	yes	D	3	0	0
2	0.3	32	no	D	0	2	0
3	0.1	128	yes	D	1	0	1
4	0.1	64	no	D	2	2	0
5	0.03	32	no	D	0	2	3
6	0.01	32	no	D	0	1	2
7	0.03	256	no	C	1	0	2
8	0.03	64	yes	C	2	1	2
9	0.01	128	yes	C	0	1	0
10	0.01	32	yes	C	1	2	0
11	0.001	64	no	C	2	1	0
12	0.001	256	yes	C	0	0	2
draws in the sample (row sums)					12	12	12

Each cell: how many times the sample drew that run (darker = more).

S1 missed runs 2, 5, 6, 9, 12.
S2 missed 1, 3, 7, 12.
S3 missed 1, 2, 4, 9, 10, 11.

A sample holds about 63% of the distinct runs: a run is missed with probability $(1 - \frac{1}{12})^{12} = 0.352$ ($\to 1/e = 0.368$).

Slide 24 — $(1 - \frac{1}{12})^{12} = 0.352 \to 1/e$ (A5).

Slide 25 Step 2: grow a tree on S1 and S2

Step 2: grow a tree on each sample (S1 and S2)

S1: 6 **D** draws (runs 1,1,1,3,4,4), 6 **C** draws (7,8,8,10,11,11), $H = 1$. **Rate** ≤ 0.065 splits them perfectly:

$$\text{gain} = 1 - \left(\tfrac{6}{12} \cdot 0 + \tfrac{6}{12} \cdot 0\right) = 1.0000 \quad \Rightarrow \text{ tree 1 stops after one question.}$$

S2: 7 **D** draws (2,2,4,4,5,5,6), 5 **C** draws (8,9,10,10,11), $H(\tfrac{7}{12}) = 0.9799$. Best root: **warm-up** = **no** (8 draws, 7 **D**, $H = 0.5436$; and 4 draws, all **C**):

$$\text{gain} = 0.9799 - \left(\tfrac{8}{12} \times 0.5436 + \tfrac{4}{12} \times 0\right) = 0.9799 - 0.3624 = 0.6175$$

The 8-draw group holds run 11 (**C**), so tree 2 asks **rate** ≤ 0.0055 to isolate it (gain 0.5436).
S2 lacks runs 1 and 3 (the **D** runs *with* warm-up), so warm-up looks like the cause.

Slide 25 — S2 lacks runs 1 and 3, so warm-up looks like the cause (A6 Step 2).

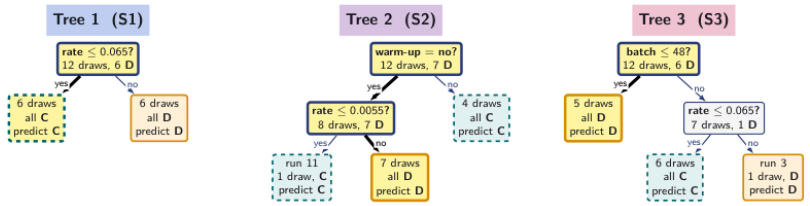
Slide 26 Step 2: tree 3, and the three trees

Shown in A6. S3: batch ≤ 48 gain 0.6549, then rate ≤ 0.065 isolates run 3.

Slides 27–28 Step 3: predictions of runs A and B by each tree

Step 3: prediction of run A by each tree

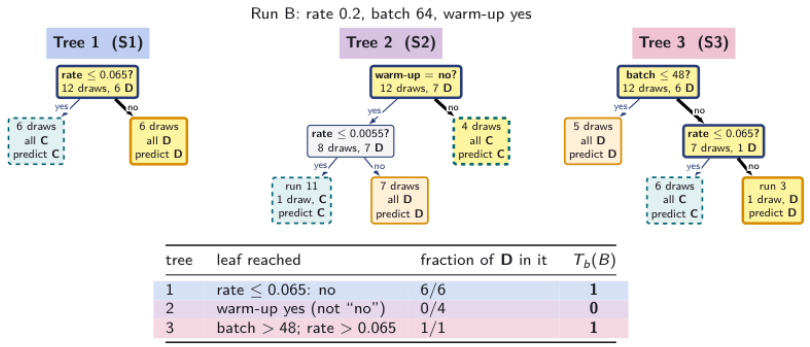
Run A: rate 0.05, batch 32, warm-up no



tree	leaf reached	fraction of **D** in it	$T_b(A)$
1	rate ≤ 0.065: yes	0/6	**0**
2	warm-up no; rate > 0.0055	7/7	**1**
3	batch ≤ 48: yes	5/5	**1**

Slide 27 — $T(A) = 0, 1, 1$.

Step 3: prediction of run B by each tree



tree	leaf reached	fraction of **D** in it	$T_b(B)$
1	rate ≤ 0.065: no	6/6	**1**
2	warm-up yes (not "no")	0/4	**0**
3	batch > 48; rate > 0.065	1/1	**1**

Slide 28 — $T(B) = 1, 0, 1$. Tree 2 says C because its sample has no high-rate run with warm-up.

Slide 29

Step 4: average the numbers, threshold at ½

Step 4: average the numbers, threshold at $\frac{1}{2}$

$\bar{T} = \frac{1}{B}\sum_b T_b$; the bagged answer is **D** if $\bar{T} > \frac{1}{2}$.

run A: $\bar{T} = \frac{1}{3}(0 + 1 + 1) = \frac{2}{3} \Rightarrow$ **D** run B: $\bar{T} = \frac{1}{3}(1 + 0 + 1) = \frac{2}{3} \Rightarrow$ **D**

Same answer as the single tree, but the single tree showed no doubt; the vote shows one tree in three disagrees.

Three trees give a rough estimate

With 20,000 bootstrap trees (simulation), the fraction saying **D** settles at 0.872 at run A and 0.934 at run B. With 200 trees: 0.845 and 0.965.

Slide 29 — The bag shows the disagreement the single tree hid.

Slides 30–31

Your turn: average the trees (and answer)

Answer: average the trees

1. $T_1 = \frac{3}{4} = 0.75,\ T_2 = 0,\ T_3 = 1.$

2. $\bar{T} = \frac{1}{3}(0.75 + 0 + 1) = 0.583 > \frac{1}{2}$: predict **D**.

3. Hard votes: tree 1 says **D**, tree 2 says **C**, tree 3 says **D**, so the share for **D** is $\frac{2}{3} = 0.667$. Both rules predict **D**, but with different confidence (0.583 against 0.667). Averaging leaf fractions is called soft voting; counting class votes is hard voting.

Slide 31 — Averaging leaf fractions = **soft voting**; counting class votes = **hard voting** (A7).

Slide 32

What σ² and ρ measure

What σ^2 and ρ measure

Fix one new input, for example run A. Each bagged tree b gives a number T_b at A. To keep it simple, think of $T_b = 1$ if the tree says **D** and 0 if it says **C**.

- $\sigma^2 = \mathrm{Var}(T_b)$ is **how much one tree's answer at A varies from tree to tree**. It is 0 if every tree gives the same answer, and largest when the trees split evenly.
- ρ is **how much two different trees' answers move together**. $\rho = 0$: one tree's answer tells you nothing about another's. $\rho = 1$: the two trees always agree.
- $\mathrm{Cov}(T_b, T_c) = \rho\sigma^2$ combines the two.

share of trees saying **D** at the input, p	$\sigma^2 = p(1-p)$	meaning
1.00	0	every tree agrees: nothing to average away
0.872 (run A, simulation on the next slides)	0.1115	most trees agree, a few differ
0.50	0.25	trees split evenly: one tree is no better than a guess

σ^2 describes one tree at one input. Averaging B trees is useful because it shrinks this spread.

Slide 32 — σ² describes one tree at one input; averaging B trees shrinks it (A4).

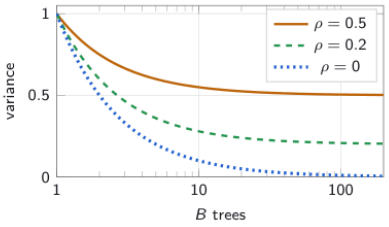
Slides 33–34 Why averaging helps: the variance (1 and 2 of 2)

Why averaging helps: the variance (2 of 2)

4. **Divide by B^2:**

$$\mathrm{Var}(\bar{T}) = \frac{\sigma^2}{B} + \frac{B-1}{B}\rho\sigma^2 = \boxed{\rho\sigma^2 + (1-\rho)\frac{\sigma^2}{B}}$$

5. **Read it:** $\rho = 0$: σ^2/B. $\rho = 1$: σ^2, no gain. $B \to \infty$: floor $\rho\sigma^2$.



$\sigma^2 = 1$: with $\rho = 0.5$, $B = 10$ gives 0.55, $B = 100$ gives 0.505.

Slide 34 — Five-step derivation in A4. σ² = 1, ρ = 0.5: B = 10 → 0.55; B = 100 → 0.505.

Slides 35–36 Your turn: the variance of an average (and answer)

σ² = 0.8, ρ = 0.25. (1) B = 16: $0.2 + 0.0375 = 0.2375$. (2) Floor 0.2. (3) $0.6/B \leq 0.01 \Rightarrow B \geq 60$; 120 trees only 0.21 → 0.205.

Slide 37 Checking the variance formula by simulation

Checking the variance formula by simulation

Resample the twelve runs, grow a tree, record its prediction at run A, repeat. At A a tree says **D** with probability $p = 0.872$, so $\sigma^2_{\text{boot}} = p(1-p) = 0.1115$.

B trees averaged	measured variance of the average	σ^2_{boot}/B
1	0.1118	0.1115
3	0.0362	0.0372
10	0.0111	0.0111
50	0.0021	0.0022

They agree: this confirms the σ^2/B part for trees grown from the *same* twelve runs. A floor does not show here, because only resampling varies; the floor comes from the dataset itself varying.

Slide 37 — Confirms σ²/B; no floor because only resampling varies (A9).

Slide 38 Where the correlation ρ comes from

Where the correlation ρ comes from

Given the data D, bootstrap trees are independent. Write $\mu(D) = E[T \mid D]$. For $b \neq c$:

$$\mathrm{Cov}(T_b, T_c) = \underbrace{E[\mathrm{Cov}(T_b, T_c \mid D)]}_{=0} + \mathrm{Var}(\mu(D)), \qquad \sigma^2 = \mathrm{Var}(\mu(D)) + E[\mathrm{Var}(T \mid D)]$$

$$\rho\sigma^2 = \mathrm{Var}(\mu(D)) \ \text{(part due to the data)}, \qquad (1-\rho)\sigma^2 = E[\mathrm{Var}(T \mid D)] \ \text{(part due to resampling)}$$

In words

Averaging removes the variation due to resampling but not the variation of the dataset itself: that is the floor. On a simulated 200-run example, $\sigma^2 = 0.071$, $\rho = 0.26$, floor 0.019; a single full tree's validation error 0.144 fell to 0.115 for a 200-tree bag.

Slide 38 — Averaging removes the resampling part, not the data part.

Slide 39 Feature use across bagged trees; random forests

Feature use across bagged trees; random forests

No: in plain bagging every split may use any feature. Root-question feature over 20,000 bootstrap trees:

	rate	batch	warm-up
bagging (all 3 features at every split)	80.8%	12.2%	6.9%
random forest, 2 random features per split	58.7%	28.0%	13.2%
random forest, 1 random feature per split	35.2%	33.2%	31.6%

Plain bagging starts with the learning rate in about four trees of five: trees alike, ρ high, floor high. A **random forest** shows each split only some features, so trees differ more, ρ falls, and so does the floor (simulated: ρ 0.26 $\rightarrow$ 0.20, floor 0.019 $\rightarrow$ 0.016), at the price of each tree being a little worse.

Slide 39 — Random forest: ρ 0.26 → 0.20, floor 0.019 → 0.016 (A10).

Slide 40 Out-of-bag error

Out-of-bag error

run	true	trees not drawing it	frac. **D**	OOB	right?
1	**D**	63	0.746	**D**	yes
2	**D**	65	0.985	**D**	yes
3	**D**	68	0.632	**D**	yes
4	**D**	77	0.818	**D**	yes
5	**D**	70	0.557	**D**	yes
6	**D**	74	0.189	**C**	**no**
7	**C**	69	0.406	**C**	yes
8	**C**	66	0.455	**C**	yes
9	**C**	80	0.037	**C**	yes
10	**C**	69	0.783	**D**	**no**
11	**C**	70	0.514	**D**	**no**
12	**C**	68	0.000	**C**	yes

A run left out of a tree's sample is a fresh test run for that tree. With $B = 200$, each run was out of bag in about 70 trees.

OOB error: 3 wrong of 12 (0.25). The single tree scored on its own runs: 0 errors, an optimistic estimate.

No runs need to be held out.

Slide 40 — OOB error 0.25; single tree on its own runs: 0 (optimistic). No runs need to be held out (A11).

Slide 41 Bagging: summary of the procedure

Bagging: summary of the procedure

step	what you do	which number is used
1. resample	draw 12 runs with replacement, B times	draw counts per run
2. grow	full tree on each sample	each tree's own questions and leaves
3. ask	send the new run down every tree	T_1, T_2, T_3: the **D**-fraction of the leaf reached
4. average	$\bar{T} = \frac{1}{B}\sum_b T_b$; **D** if $\bar{T} > \frac{1}{2}$	the numbers, added and divided by B
why it works	$\mathrm{Var}(\bar{T}) = \rho\sigma^2 + (1-\rho)\sigma^2/B$	σ^2: one tree; ρ: shared data
cannot fix	the floor $\rho\sigma^2$, and the bias	lower ρ: random forest

Slide 41 — Cannot fix: the floor ρσ² and the bias. Lower ρ: random forest.

Part B′ — The Notes (L12c and L12d)

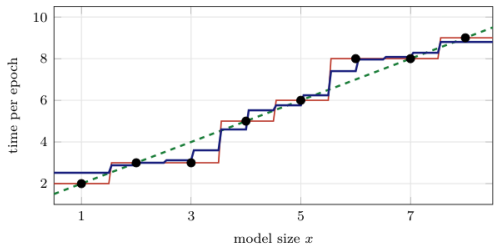
Notes L12c, §1–2 — a tree that memorises; what variance means

Eight points from $\mu(x) = x + 1$; a full tree is a nearest-neighbour step function with training error 0 because it fitted the noise. Prediction variance, bias and the six-set table: A1. Your turn 1: at $x_0 = 2.7$ and 3.4 the nearest point is 3 ⇒ prediction 3; truths 3.7 and 4.4 ⇒ errors 0.7 and 1.4 (the noise –1 at $x = 3$ is copied over a longer stretch). Your turn 2: predictions 2.9, 5.1, 4.2, 3.4, 5.4 ⇒ mean 4.20, squared deviations sum 4.58 ⇒ variance 0.916.

Notes L12c, §3–4 — averaging and the bootstrap

A3–A5. Your turn 3: σ² = 0.25, B = 4: (a) independent 0.0625; (b) ρ = 0.5: $0.125 + 0.03125 = 0.15625$ — four correlated trees are about as noisy as two independent ones. Your turn 4: $(7/8)^8$: $0.7656 \to 0.5862 \to 0.3436$.

Notes L12c, §5 — bagging step by step



Notes Fig. 5 — The blue (bagged) curve is smoother and nearer the truth.
Your turn 5: a sixth sample with $x = 3, \ldots, 8$ ⇒ nearest to 4.6 is 5 ⇒ 6.0;
average of six $= 35/6 = 5.83$.

Notes L12c, §6 — bias and variance: what each method fixes

Table and noise-free boosting example in A12. ρ measured from pairs (A9 Step 5). Your turn 6: $u = 4, 6, 8, 5, 7$, $v = 5, 5, 7, 6, 7$: means 6, 6; covariance $5/5 = 1$; var(u) = 2, var(v) = 0.8 ⇒ $r = 1/\sqrt{1.6} = 0.79$.

Notes L12c, §7–8 — voting; random forests

Majority-vote table and two warnings: A7. Random forest: at every split look at a random subset (e.g. $\sqrt{d}$ of d) ⇒ ρ falls, each tree a bit weaker; in practice the lower correlation wins.

Notes L12c, §9 — derive-it-yourself boxes (summary)

D1: bias² + variance + noise (A2). D2: σ²/B (A3). D3: ρσ² + (1−ρ)σ²/B (A4). D4: $(1 - 1/N)^N \to e^{-1}$ (A5). D5: $E[\bar{f}] = m(x)$ ⇒ averaging leaves bias alone (A3 Step 4).

Notes L12d — bagging on decision trees (worked example)

The same computation as deck slides 23–41, written out as a document: the single tree, three bootstrap samples, three trees, paths of A and B, the variance derivation, where ρ comes from, root-feature shares, OOB table, and five review questions (answers: S2 lacks runs 1 and 3; $T_2(B) = 0$; $0.3 + 0.7/20 = 0.335$, floor 0.3; no floor because only resampling varies; random forest hides features so trees use different questions). All covered in A6–A11.

Also worth knowing

Why a committee beats its members. If B independent voters are each right with probability $p > \frac{1}{2}$, the probability that the majority is right climbs towards 1 as B grows. With $p < \frac{1}{2}$ it falls towards 0, so a committee needs members better than chance and errors that are not all the same.

Robustness. Averaging makes bagging tolerant of noise and label errors, because a bad point appears in only about 63% of the samples and is outvoted. Boosting is more exposed, since it keeps concentrating on the points it finds hard, and mislabeled points are hard for ever.

Part C Practice

From the practice sheet (`L12g` , Parts B–D) plus deck review questions.

Formulas: P(run missed) $= (1 - \frac{1}{n})^n \to e^{-1}$; soft vote = mean of leaf fractions; hard vote = share of trees with fraction > ½; $\mathrm{Var}(\bar{T}) = \rho\sigma^2 + (1-\rho)\sigma^2/B$, floor $\rho\sigma^2$.

Part B — Bootstrap and bagging

1. **(P7)** (a) $n = 12$: P(given run not drawn), and expected number of distinct runs. (b) Compare $n = 100$, $n \to \infty$. (c) P(run 3 drawn exactly twice), $n = 12$.

2. **(P8)** Sample $S = (1, 1, 3, 4, 4, 6, 7, 7, 7, 9, 10, 12)$. (a) Out-of-bag runs? (b) Distinct runs seen? (c) Most-drawn run?

3. **(P9)** Five trees' leaf spam fractions: 0.55, 0.52, 0.51, 0.05, 0.02. (a) Soft vote? (b) Hard vote? (c) Why do they differ?

4. **(P10)** Out-of-bag votes for 8 training emails:

email	1	2	3	4	5	6	7	8
true	spam	spam	not	not	spam	not	spam	not
trees not drawing it	3	4	3	5	2	4	3	3
of these, voting spam	2	1	0	3	2	1	3	1

   (a) OOB prediction for emails 2–8 (email 1: 2/3 spam ⇒ spam, correct). (b) OOB error. (c) Why is a full tree's 0 training mistakes not a good estimate, while OOB is?

Part C — Variance of an average (background: from $K = 6$ repeats of two trees, σ² ≈ 0.5 and ρ ≈ 0.3; check: average of two trees has variance 0.322 = $0.146 + 0.176$ ✓; B = 4 gives $\frac{1}{16}(4 \times 0.5 + 12 \times 0.15) = 0.2375$.)

5. **(P11)** σ² = 0.5, ρ = 0.3. Variance of the average of (a) 10, (b) 100, (c) ∞ trees; reduction factor each time.
6. **(P12)** σ² = 0.4, B = 25. Variance when ρ = 0, 0.5, 1. What does each say about averaging?
7. **(P13)** σ² = 1, ρ = 0.2. (a) Trees to get within 0.01 of the floor? (b) Trees for variance ≤ 0.25? (c) Can it reach 0.15?

8. **(P14)** Single tree variance 0.12; average of B = 10 has 0.048. Find ρ and the floor.
9. **(P15)** Six trees' leaf fractions at one input: 0.9, 0.7, 0.8, 0.4, 1.0, 0.6. (a) Prediction? (b) σ² by sample variance (divide by B – 1). (c) What does σ² measure here?

Part D — Random forests 10. **(P16)** $p = 5$ features, $m = 2$ offered per split. (a) Number of pairs? (b) P(strongest feature offered)? (c) P(two specific features both offered)? (d) $p = 100$, $m = 10$: P(strongest offered)? (e) With $m = 10$ and 15 splits, P(strongest never offered)? 11. **(P17)** Bagging: rate is root in 80.8% of trees. Random forest offering 2 of 3: rate wins 88% of trees where offered. (a) Fraction of trees where rate is offered at the root? (b) Where rate is the root? (c) Why is (b) < 80.8%, and why does it lower ρ? 12. **(P18)** Bagging σ² = 0.071, ρ = 0.26; forest σ² = 0.080, ρ = 0.20; B = 200. (a) Variance of each average. (b) Which is lower? (c) If the forest's σ² were 0.11, does it still win? 13. **(Deck review)** σ² = 1, ρ = 0.3, B = 20: variance and floor? Which number do we read from tree 2 for run B, and why?

1. (a) $(11/12)^{12} = 0.3520$; drawn with prob. 0.6480 ⇒ expected distinct $12 \times 0.648 = 7.78$. (b) $0.99^{100} = 0.3660$; $e^{-1} = 0.3679$ — about a third left out regardless of n. (c) Binomial(12, 1/12): $\binom{12}{2}(\frac{1}{12})^2(\frac{11}{12})^{10} = 66 \times 0.006944 \times 0.4189 = 0.192$.
2. Distinct {1, 3, 4, 6, 7, 9, 10, 12}. (a) OOB {2, 5, 8, 11}. (b) 8 distinct (4/12 = 0.333 left out, close to 0.352). (c) Run 7, three times (1 and 4 twice; total 12 draws ✓).
3. (a) Mean $1.65/5 = 0.33$ ⇒ not spam. (b) 3 of 5 trees > 0.5 ⇒ 0.6 ⇒ spam. (c) Soft uses confidence: two very confident trees (0.05, 0.02) pull hard; hard voting counts every tree once, so three borderline trees win. Soft is the usual default for probabilities.
4. (a) Shares: 2/3, 1/4, 0/3, 3/5, 2/2, 1/4, 3/3, 1/3 ⇒ spam, not, not, spam, spam, not, spam, not. Correct: 1, 3, 5, 6, 7, 8; wrong: **2 and 4**. (b) $2/8 = 0.25$. (c) A full tree has seen and memorised its training emails; for OOB each email is predicted only by trees that never saw it, so it acts as fresh test data — and nothing is held out.
5. ρσ² = 0.15, (1–ρ)σ² = 0.35. (a) $0.15 + 0.035 = 0.185$ (2.7× smaller). (b) 0.1535 (3.3×). (c) 0.15 (at best 3.3×). Past the floor, more trees stop helping.
6. (a) $0.4/25 = 0.016$ (25× smaller). (b) $0.2 + 0.008 = 0.208$ (about 2×). (c) 0.4 (nothing). Averaging helps only to the extent that the trees differ.
7. Floor 0.2, resampling part 0.8/B. (a) $0.8/B \le 0.01 \Rightarrow B \ge 80$. (b) $0.2 + 0.8/B \le 0.25 \Rightarrow B \ge 16$. (c) No: always > 0.2. Lower ρ (random forest) to go below.
8. $0.048 = 0.12\rho + 0.012(1 - \rho) = 0.012 + 0.108\rho \Rightarrow \rho = 1/3$; floor $= 0.04$. Check: $0.04 + 0.667 \times 0.012 = 0.048$ ✓.
9. (a) $4.4/6 = 0.733 > 0.5$ ⇒ spam. (b) Deviations +0.167, −0.033, +0.067, −0.333, +0.267, −0.133; squares sum 0.2333; $/5 = 0.0467$. (c) The tree-to-tree spread of one tree's answer at this fixed

input; the six trees are six draws from that variation.

10. (a) $\binom{5}{2} = 10$. (b) $4/10 = m/p = 0.4$. (c) $1/10$. (d) $10/100 = 0.1$. (e) $0.9^{15} = 0.206$ — about one tree in five never gets to use the strongest feature: exactly what makes forest trees differ.
11. (a) Pairs {rate, batch}, {rate, warm-up} of 3 ⇒ $\frac{2}{3}$. (b) $\frac{2}{3} \times 0.88 = 0.587$. (c) Rate is hidden from a third of roots, so batch and warm-up start more trees (28.0%, 13.2%); different starts ⇒ different trees ⇒ lower ρ.
12. (a) Bagging: $0.26 \times 0.071 = 0.0185$ + $0.74 \times 0.071/200 = 0.00026$ ⇒ **0.0187**. Forest: $0.016 + 0.00032$ ⇒ **0.0163**. (b) Forest: the drop in ρ outweighs the rise in σ². (c) $0.022 + 0.00044 = 0.0224 > 0.0187$ ⇒ bagging wins. Tune m on held-out data or OOB error.
13. $0.3 + 0.7/20 = 0.335$; floor 0.3. $T_2(B) = 0$: warm-up is yes, so B goes to the leaf "4 draws, all C".

Next: L12c — Gradient boosting: instead of averaging independent deep trees, add up small trees, each fitted to what the previous ones got wrong.